NILAMBARA ELECTRICALS

System test report

Repair Shop Management System

Version 1.0 | 20 October 2026

Step 12 follow-up handover

Software verification complete for the recorded test scope.
Business acceptance and operational configuration items remain pending.

System test report

Executive result and evidence date

The final isolated local verification on 20 October 2026 passed **218 tests with 3566 assertions in 175.11 seconds**, plus native multi-process concurrency and actual disposable database dump/restore checks. This report records the executed software checks. Business acceptance remains pending and no production readiness, exhaustive penetration testing or capacity certification is claimed.

Reading the consolidated evidence

The following chapters reproduce the maintained test reports, with whitespace normalized. Some chapters retain original Step 12 findings followed by a Pending-work follow-up. The follow-up and the first completion chapter supersede earlier statements that stock/supplier workflows were absent or native database checks were still pending. This preserves traceability rather than erasing the audit history. UAT manual statuses remain pending even when automated coverage exists.

The authoritative execution log is storage/logs/pending-complete-mysql.log. Schema snapshots, preservation, syntax, style and PDF evidence are referenced by the completion chapter. The full requirement-by-requirement matrix is docs/testing/brd-compliance-matrix.md and the method inventory is docs/testing/automated-test-inventory.md in the delivered repository; these are detailed companion evidence rather than assertions that every BRD requirement is accepted.

Environment and safety

Laravel 13 / PHP 8.5, the MySQL driver against local XAMPP MariaDB 10.4.32, InnoDB/utf8mb4. The helper created a randomized local disposable test database, exercised migrations/recovery there and removed only that fixture database. Operational tables were not reset or migrated during this verification. Matching before/after row counts do not prove byte-identical operational records.

Acceptance still required

Approved operational backup destination; actual Google Drive configuration/upload; physical printer/scanner tests; realistic-volume capacity acceptance; owner UAT/signatures; purchase-wide discount/freight costing decision. No live operational restore, production build, scheduler activation, installer or deployment was performed.

Pending-work completion report

The missing operational modules have now been implemented and tested. Full business acceptance remains conditional on the external items below; this is not a 100% BRD-compliance declaration.

Final native MySQL result: **218 passed (3566 assertions)**, duration 175.11s. Full evidence: `storage/logs/pending-complete-mysql.log`. The helper created its own randomized database, ran tests, rolled back/remigrated that disposable database, ran native multiuser/recovery checks, then removed only that database. Operational tables were not migrated/reset.

Focused evidence: `pending-focused-final.log`; PDF dimensions/fonts: `pending-pdf-review.json`; before/after operational schema/count evidence: `pending-schema-before.json`, `pending-schema-after.json`, `pending-preservation.log`; syntax/style evidence: `pending-syntax.log`, `pending-style-check.log`. All names are under `storage/logs/`.

Implemented: supplier registration/edit/search and opening balances; inventory master, references, preferred suppliers and optional known opening costs; purchase drafts/finalization, partial GRNs and source-linked stock receiving; authorized repair requests/issues and complete unused-issue returns; corrections/damage/counts; supplier returns with explicitly agreed credit; supplier purchase/opening payments and reversals; purchase/GRN print/PDF/history; receipt/card PDF export; statement template/paper/Sinhala/history fixes; approved quotation charge transfer; unquoted labour charge entry; completion/delivery timestamps; latest-purchase valuation and recorded-cost profit estimates.

Financial corrections: accounting payment reversals reduce collections without reducing an issued sale; real refunds remain separate. Older returned goods preserve their original cost but cannot replace a newer purchase cost. Repair quote approval transfers selling charges without posting stock or payments. Financial and stock writes are transactional, permission checked, and retain original documents.

An approved live-backup folder has not been supplied; Google Drive folder/account/OAuth fields remain blank as requested. No host/production scheduler, production build, installer, packaging or deployment was performed. Physical printer/scanner checks, realistic-volume capacity acceptance and owner signatures remain pending. Latest purchase cost was explicitly selected by the owner. Purchase-wide discount/freight allocation remains a separate unanswered decision; affected profit estimates remain incomplete rather than assuming a policy. Existing role permissions were preserved: storekeepers can receive/issue/count/adjust, while purchase approval and supplier money operations require `purchases.approve`.

Native checks

- PASS: two cashiers buying four of five units; one sale, one rejection, one unit remains.
- PASS: simultaneous independent checkouts receive distinct sale numbers.
- PASS: simultaneous identical retries create one sale and one payment.
- PASS: two simultaneous cashiers, invoice, one save and one rejection; barrier-to-completion 165.828 ms.

- PASS: two simultaneous cashiers, payment, one save and one rejection; barrier-to-completion 198.91 ms.
- PASS: one invoice-job reservation and exactly 100.00 allocated, no overpayment.
- PASS: service/PDF observations recorded: {"invoice-create-and-issue":57.288,"payment-receive-and-allocate":63.399,"pos-checkout-including-stock-posting":34.648,"a5-invoice-pdf-with-print-history":1424.018}
- PASS: two receptionists create atomic receipts and unique jobs; 2 concurrent workers; 149.996 ms; ["SAVED:1","SAVED:2"]
- PASS: two technicians update their assigned jobs with retained histories; 2 concurrent workers; 103.372 ms; ["SAVED:5","SAVED:6"]
- PASS: POS and repair issue compete for the same five units; 2 concurrent workers; 163.011 ms; ["SAVED:1","REJECTED:{"cart":["Insufficient saleable stock for Test spare part. Available: 1.000."]}"]
- PASS: two storekeepers cannot over-receive a purchase; 2 concurrent workers; 151.152 ms; ["REJECTED:{"quantity":["Quantity exceeds the unreceived purchase quantity."]}","SAVED:1"]
- PASS: receipt, technician, shared stock and GRN race invariants verified in disposable MySQL.
- PASS: real MySQL dump created, nonempty, SHA-256 verified, private and protected.
- PASS: confirmation and wrong-password refusals; authenticated CLI restore, safety dump, maintenance lifecycle, data recovery and audit history.

Practical boundaries

Repair return currently reverses an entire unused, resalable issue before billing; damaged goods use the damage workflow. Receiving posts one purchase line/location per GRN submission, supporting partial deliveries. Supplier returns for historical purchases containing repeated lines for the same part are refused until their source is reconciled. No financial records are deleted to simplify these cases. Profit is an estimate of recorded costs, not an audited net-income certification.

Integration test report

Final native MySQL result: **218 passed (3566 assertions)**, duration 175.11s. Full evidence: `storage/logs/pending-complete-mysql.log`. The helper created its own randomized database, ran tests, rolled back/remigrated that disposable database, ran native multiuser/recovery checks, then removed only that database. Operational tables were not migrated/reset.

Focused evidence: `pending-focused-final.log`; PDF dimensions/fonts: `pending-pdf-review.json`; before/after operational schema/count evidence: `pending-schema-before.json`, `pending-schema-after.json`, `pending-preservation.log`; syntax/style evidence: `pending-syntax.log`, `pending-style-check.log`. All names are under `storage/logs/`.

Verified chains: intake/advance → approval pricing → repair stock issue → invoice → net payments → delivery; supplier purchase draft → approval/payable → partial receiving → stock ledger → supplier payment → reversal; supplier return → stock deduction → explicitly agreed supplier credit; POS versus repair competing for the same stock. Existing customer financial tables are untouched by supplier payment operations.

Draft purchase, quotation and part request saves do not change stock. Approval sets job prices but does not issue parts or collect money. Repeat posts are protected by stable keys and locked source documents. Failure cases and the native races exercise duplicate/over-receiving/overselling protection.

Native concurrency covers two receptionists, two assigned technicians on their respective jobs, two invoice/payment cashiers, POS versus repair issue, and two GRN receivers. This does not establish a whole-shop capacity limit or same-record optimistic conflict detection for arbitrary edits.

- PASS: two cashiers buying four of five units; one sale, one rejection, one unit remains.
- PASS: simultaneous independent checkouts receive distinct sale numbers.
- PASS: simultaneous identical retries create one sale and one payment.
- PASS: two simultaneous cashiers, invoice, one save and one rejection; barrier-to-completion 165.828 ms.
- PASS: two simultaneous cashiers, payment, one save and one rejection; barrier-to-completion 198.91 ms.
- PASS: one invoice-job reservation and exactly 100.00 allocated, no overpayment.
- PASS: service/PDF observations recorded: {"invoice-create-and-issue":57.288,"payment-receive-and-allocate":63.399,"pos-checkout-including-stock-posting":34.648,"a5-invoice-pdf-with-print-history":1424.018}
- PASS: two receptionists create atomic receipts and unique jobs; 2 concurrent workers; 149.996 ms; ["SAVED:1","SAVED:2"]
- PASS: two technicians update their assigned jobs with retained histories; 2 concurrent workers; 103.372 ms; ["SAVED:5","SAVED:6"]
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- PASS: two storekeepers cannot over-receive a purchase; 2 concurrent workers; 151.152 ms; ["REJECTED:{"quantity":["Quantity exceeds the unreceived purchase quantity."]}","SAVED:1"]
- PASS: receipt, technician, shared stock and GRN race invariants verified in disposable MySQL.
- PASS: real MySQL dump created, nonempty, SHA-256 verified, private and protected.
- PASS: confirmation and wrong-password refusals; authenticated CLI restore, safety dump, maintenance lifecycle, data recovery and audit history.

An approved live-backup folder has not been supplied; Google Drive folder/account/OAuth fields remain blank as requested. No host/production scheduler, production build, installer, packaging or deployment was performed. Physical printer/scanner checks, realistic-volume capacity acceptance and owner signatures remain pending. Latest purchase cost was explicitly selected by the owner. Purchase-wide discount/freight allocation remains a separate unanswered decision; affected profit estimates remain incomplete rather than assuming a policy. Existing role permissions were preserved: storekeepers can receive/issue/count/adjust, while purchase approval and supplier money operations require `purchases.approve`.

Functional test report

Final native MySQL result: **218 passed (3566 assertions)**, duration 175.11s. Full evidence: `storage/logs/pending-complete-mysql.log`. The helper created its own randomized database, ran tests, rolled back/remigrated that disposable database, ran native multiuser/recovery checks, then removed only that database. Operational tables were not migrated/reset.

Focused evidence: `pending-focused-final.log`; PDF dimensions/fonts: `pending-pdf-review.json`; before/after operational schema/count evidence: `pending-schema-before.json`, `pending-schema-after.json`, `pending-preservation.log`; syntax/style evidence: `pending-syntax.log`, `pending-style-check.log`. All names are under `storage/logs/`.

The final suite exercises the prior customer, intake, quotation, invoice, payment, POS, role/security/report/backup functionality and the new stock/purchasing/supplier workflows. StockPurchasingWorkflowTest covers the actual customer advance → receipt/job → quotation approval → stock issue → invoice → remaining payment → delivery path without directly setting billable job amounts. Inspection/technician authorization is also covered by the existing job suite and native assigned-job tests. Supplier edit/search preserves opening financial history. Preferred-supplier/opening-cost inputs are verified.

LatestPurchaseCostTest verifies newer purchase cost survives an older returned item's movement, exact valuation and refund-aware recorded margin; payment correction does not become a sales refund. PrintGapCompletionTest verifies supplier/GRN documents and statement settings/history. The complete source method inventory is in automated-test-inventory.md. A passing suite does not replace physical or owner UAT. An approved live-backup folder has not been supplied; Google Drive folder/account/OAuth fields remain blank as requested. No host/production scheduler, production build, installer, packaging or deployment was performed. Physical printer/scanner checks, realistic-volume capacity acceptance and owner signatures remain pending. Latest purchase cost was explicitly selected by the owner. Purchase-wide discount/freight allocation remains a separate unanswered decision; affected profit estimates remain incomplete rather than assuming a policy. Existing role permissions were preserved: storekeepers can receive/issue/count/adjust, while purchase approval and supplier money operations require `purchases.approve`.

Database relationship audit

Read-only audit of `nilambara_repair_management` on 2026-10-20T09:41:16+05:30. No application migrations or business data writes were performed by this audit script.

- 69 tables inspected, 247 foreign-key column mappings inspected, 0 orphan matches detected by per-column checks.
- Non-InnoDB/non-utf8mb4 tables: none.
- Floating-point storage columns: none.
- Compound associations are additionally verified by BusinessConstraintsTest and BusinessRelationshipsTest in the disposable database. The per-column orphan scan alone does not prove compound-key integrity.
- Infrastructure tables may use string primary keys. No primary key/index changes were made.
- Operational migration state: `storage/logs/step12-migrations.log`; route inventory: `storage/logs/step12-routes.json`.
- Full PK/FK ERD: existing ERD. No schema changes in Step 12, so no ERD relationship changes are necessary.

Tables and indexes

Table	Rows at read-only snapshot	Index names
audit_logs	16	PRIMARY, ix_audit_logs_da9eab2f, ix_audit_logs_92da6dde, ix_audit_logs_5fe56ab1, ix_audit_logs_cace4a15, audit_logs_occurred_at_index
backup_histories	0	PRIMARY, ix_backup_histories_5af47fbb, ix_backup_histories_48a3661d, ix_backup_histories_5fb88229
backup_restores	0	PRIMARY, ix_backup_restores_b9e34ae4, ix_backup_restores_5fb88229
cache	0	PRIMARY, cache_expiration_index
cache_locks	0	PRIMARY, cache_locks_expiration_index
customers	0	PRIMARY, uq_customers_68db6340, ix_customers_ac145587, ix_customers_a76c5cd4, ix_customers_e7e1e7cc

Table	Rows at read-only snapshot	Index names
customer_account_entries	0	PRIMARY, uq_customer_account_entries_a5d450da, uq_customer_account_entries_7db4ecce, uq_customer_account_entries_1de3c56e, uq_customer_account_entries_15d25c20, uq_customer_account_entries_bc1ec442, uq_customer_account_entries_fcaec7df, uq_customer_account_entries_5b312e0c, uq_customer_account_entries_fc740855, fk_customer_account_entries_5585485a, fk_customer_account_entries_62d0b35a, fk_customer_account_entries_4355babb, fk_customer_account_entries_5bcc665a, fk_customer_account_entries_afce3f4c, ix_customer_account_entries_d947b0a0, ix_customer_account_entries_a7a13f4c, ix_customer_account_entries_06129e94
document_print_histories	0	PRIMARY, ix_document_print_histories_52a2214d, ix_document_print_histories_66747a19, ix_document_print_histories_cf053e5b, ix_document_print_histories_596c572d, ix_document_print_histories_15d25c20, ix_document_print_histories_bc1ec442, ix_document_print_histories_fcaec7df, ix_document_print_histories_a7a13f4c, ix_document_print_histories_86cc2ebe, ix_document_print_histories_432b819f
document_sequences	2	PRIMARY, uq_document_sequences_aacd3103
employees	6	PRIMARY, uq_employees_348a4a98, uq_employees_cace4a15
expenses	0	PRIMARY, uq_expenses_b958d1d5, ix_expenses_4545e26c, ix_expenses_478b68a9, ix_expenses_06129e94
expense_categories	0	PRIMARY, uq_expense_categories_6ae99955
failed_jobs	0	PRIMARY, failed_jobs_uuid_unique, failed_jobs_connection_queue_failed_at_index

Table	Rows at read-only snapshot	Index names
goods_received_notes	0	PRIMARY, uq_goods_received_notes_951d6c0e, uq_goods_received_notes_da00b996, fk_goods_received_notes_3e1bac6a, ix_goods_received_notes_86cc2ebe, ix_goods_received_notes_2bc0d2ca, ix_goods_received_notes_ec48f282
goods_received_note_items	0	PRIMARY, uq_goods_received_note_items_156a1229, uq_goods_received_note_items_d9f67aa6, fk_goods_received_note_items_f23d6f0e, fk_goods_received_note_items_d8bea874, fk_goods_received_note_items_25d8e711, ix_goods_received_note_items_fc342424, ix_goods_received_note_items_86cc2ebe, ix_goods_received_note_items_e7e6cd38, ix_goods_received_note_items_10e3c1ce, ix_goods_received_note_items_876d7bdb
invoices	0	PRIMARY, uq_invoices_a5d450da, uq_invoices_60284980, ix_invoices_c0000d83, ix_invoices_feb4944b, ix_invoices_a7a13f4c, ix_invoices_782ecea6, ix_invoices_43e48017, invoices_status_due_date_index
invoice_history	0	PRIMARY, ix_invoice_history_event, ix_invoice_history_actor
invoice_items	0	PRIMARY, uq_invoice_items_d0e58f0e, fk_invoice_items_cf7ebf0c, ix_invoice_items_15d25c20, ix_invoice_items_0d3839f6, ix_invoice_items_c7f10065, ix_invoice_items_10e3c1ce
invoice_jobs	0	PRIMARY, uq_invoice_jobs_393bb4c9, uq_invoice_jobs_ec756383, uq_invoice_jobs_d2dc6852, fk_invoice_jobs_62d0b35a, fk_invoice_jobs_05167f50, ix_invoice_jobs_15d25c20, ix_invoice_jobs_a7a13f4c
jobs	0	PRIMARY, jobs_queue_index

Table	Rows at read-only snapshot	Index names
job_approvals	0	PRIMARY, ix_job_approvals_d2dc6852, ix_job_approvals_06129e94
job_batches	0	PRIMARY
job_cards	0	PRIMARY, uq_job_cards_d2dc6852, ix_job_cards_52a2214d
job_inspections	0	PRIMARY, ix_job_inspections_d2dc6852, ix_job_inspections_4a26244b, ix_job_inspections_9909258e
job_part_usages	0	PRIMARY, uq_job_part_usages_d9f67aa6, uq_job_part_usages_7db4ecce, ix_job_part_usages_d2dc6852, ix_job_part_usages_10e3c1ce, ix_job_part_usages_876d7bdb, ix_job_part_usages_1d9e5646, ix_job_part_usages_46df84f0
job_status_histories	0	PRIMARY, ix_job_status_histories_04832136, ix_job_status_histories_ee56f6a4, ix_job_status_histories_d2dc6852, ix_job_status_histories_69259fd8
migrations	26	PRIMARY
password_reset_tokens	0	PRIMARY
payments	0	PRIMARY, uq_payments_ad3551cc, uq_payments_39743c6d, uq_payments_7db4ecce, uq_payments_a5d450da, ix_payments_7c3ebe1b, ix_payments_6b26a8f9, ix_payments_a7a13f4c, ix_payments_ec48f282

Table	Rows at read-only snapshot	Index names
payment_allocations	0	PRIMARY, uq_payment_allocations_ae6c2db1, uq_payment_allocations_7db4ecce, fk_payment_allocations_5bcc665a, fk_payment_allocations_05167f50, fk_payment_allocations_62d0b35a, fk_payment_allocations_4355babb, fk_payment_allocations_5f5d56da, fk_payment_allocations_e4f5269a, ix_payment_allocations_fcaec7df, ix_payment_allocations_a7a13f4c, ix_payment_allocations_d2dc6852, ix_payment_allocations_15d25c20, ix_payment_allocations_bc1ec442, ix_payment_allocations_997c1d72
payment_allocation_releases	0	PRIMARY, uq_payment_allocation_releases_7db4ecce, fk_payment_allocation_releases_cb606553, fk_payment_allocation_releases_b1bc50fb, ix_payment_allocation_releases_c33ff68d, ix_payment_allocation_releases_fcaec7df, ix_payment_allocation_releases_5b312e0c, ix_payment_allocation_releases_e78980db
payment_reversals	0	PRIMARY, uq_payment_reversals_ae6c2db1, uq_payment_reversals_7db4ecce, uq_payment_reversals_a5d450da, fk_payment_reversals_5f5d56da, ix_payment_reversals_fcaec7df, ix_payment_reversals_a7a13f4c, ix_payment_reversals_07fdd642
permissions	98	PRIMARY, uq_permissions_e6fb0621, uq_permissions_name_guard
permission_role	216	PRIMARY, uq_permission_role_1fd58356, ix_permission_role_b36bcfe0, ix_permission_role_3b8b97af
permission_user	0	PRIMARY, uq_permission_user_6d378d26, ix_permission_user_cace4a15, ix_permission_user_3b8b97af

Table	Rows at read-only snapshot	Index names
pos_sales	0	PRIMARY, uq_pos_sales_ad3551cc, uq_pos_sales_1407226b, uq_pos_sales_a5d450da, uq_step10_pos_sale_idempotency, uq_step10_pos_payment_reference, ix_pos_sales_a8fee2cf, ix_pos_sales_a7a13f4c, ix_pos_sales_782ecea6, pos_sales_payment_verified_by_user_id_foreign
pos_sale_items	0	PRIMARY, uq_pos_sale_items_3da73ab0, uq_pos_sale_items_d9f67aa6, ix_pos_sale_items_bc1ec442, ix_pos_sale_items_10e3c1ce, ix_pos_sale_items_876d7bdb
print_templates	8	PRIMARY, uq_print_templates_1f1f8ab9
purchases	0	PRIMARY, uq_purchases_4e0db050, uq_purchases_6c59959c, uq_purchases_95419c78, ix_purchases_d9987142, ix_purchases_2bc0d2ca, ix_purchases_ae7720e8, purchases_status_due_date_index
purchase_items	0	PRIMARY, uq_purchase_items_ed918c72, uq_purchase_items_951d6c0e, uq_purchase_items_26e56352, ix_purchase_items_86cc2ebe, ix_purchase_items_10e3c1ce
quotations	0	PRIMARY, uq_quotations_4bfa8930, uq_quotations_73fda6a0, uq_quotations_b0fb8224, fk_quotations_05167f50, fk_quotations_69de8f01, ix_quotations_0bd5d101, ix_quotations_d2dc6852, ix_quotations_a7a13f4c, ix_quotations_ae7720e8, quotations_updated_by_user_id_foreign, ix_quotations_prep_approval
quotation_approvals	0	PRIMARY, ix_quotation_approvals_596c572d, ix_quotation_approvals_06129e94
quotation_history	0	PRIMARY, ix_quotation_history_event, ix_quotation_history_actor

Table	Rows at read-only snapshot	Index names
quotation_items	0	PRIMARY, uq_quotation_items_9d5c5c2c, ix_quotation_items_596c572d, ix_quotation_items_10e3c1ce
repair_jobs	0	PRIMARY, uq_repair_jobs_a5d450da, uq_repair_jobs_e9c25de9, uq_repair_jobs_8bf12fb6, ix_repair_jobs_b8aca1c7, ix_repair_jobs_67b6d081, ix_repair_jobs_c6ae1260, ix_repair_jobs_a7a13f4c, ix_repair_jobs_ec48f282, ix_repair_jobs_465640da, ix_repair_jobs_43e48017, repair_jobs_received_at_index, repair_jobs_completed_at_index, repair_jobs_delivered_at_index
repair_receipts	0	PRIMARY, uq_repair_receipts_d2dc6852, uq_repair_receipts_dc11faa3, fk_repair_receipts_05167f50, ix_repair_receipts_a7a13f4c
roles	6	PRIMARY, uq_roles_e6fb0621, uq_roles_name_guard
role_user	6	PRIMARY, uq_role_user_721adde8, ix_role_user_cace4a15, ix_role_user_b36bcfe0
sessions	7	PRIMARY, sessions_user_id_index, sessions_last_activity_index
spare_parts	0	PRIMARY, uq_spare_parts_cf559ae3, uq_spare_parts_006b53f0, ix_spare_parts_6ae99955, ix_spare_parts_a5d20765, ix_spare_parts_9c4e4a89, ix_spare_parts_4252c4be, ix_spare_parts_2bc0d2ca, ix_spare_parts_4ea168a9
spare_part_brands	0	PRIMARY, uq_spare_part_brands_6ae99955
spare_part_categories	0	PRIMARY, uq_spare_part_categories_6ae99955
stock_adjustments	0	PRIMARY, uq_stock_adjustments_4fe26025, ix_stock_adjustments_ae7720e8, ix_stock_adjustments_46df84f0

Table	Rows at read-only snapshot	Index names
stock_adjustment_items	0	PRIMARY, uq_stock_adjustment_items_1e5daea3, uq_stock_adjustment_items_d9f67aa6, ix_stock_adjustment_items_a140820a, ix_stock_adjustment_items_10e3c1ce, ix_stock_adjustment_items_876d7bdb
stock_counts	0	PRIMARY, uq_stock_counts_568cf27e, ix_stock_counts_c0622bff
stock_count_items	0	PRIMARY, uq_stock_count_items_ee7e6c5a, uq_stock_count_items_d9f67aa6, ix_stock_count_items_942c8300, ix_stock_count_items_10e3c1ce, ix_stock_count_items_876d7bdb
stock_locations	0	PRIMARY, uq_stock_locations_e6fb0621
stock_movements	0	PRIMARY, uq_stock_movements_d9f67aa6, uq_stock_movements_7db4ecce, uq_stock_movements_36e939cc, uq_stock_movements_59d7f54b, uq_stock_movements_53950c4b, uq_stock_movements_eba31e8f, uq_stock_movements_81078615, uq_stock_movements_ec123f6e, uq_stock_movements_28c2bb76, fk_stock_movements_d942adbc, fk_stock_movements_037ecddd, fk_stock_movements_cc6702aa, fk_stock_movements_ad495a05, fk_stock_movements_0d8111ca, fk_stock_movements_f4ec6a20, fk_stock_movements_693404d5, ix_stock_movements_fa634624, ix_stock_movements_d084c0da, ix_stock_movements_10e3c1ce, ix_stock_movements_876d7bdb, ix_stock_movements_06129e94, ix_stock_movements_35647365
stock_returns	0	PRIMARY, uq_stock_returns_855a7dba, uq_step10_pos_return_idempotency, ix_stock_returns_d2dc6852, ix_stock_returns_bc1ec442, ix_stock_returns_86cc2ebe, ix_stock_returns_06129e94, stock_returns_refund_payment_reversal_id_foreign

Table	Rows at read-only snapshot	Index names
stock_return_items	0	PRIMARY, uq_stock_return_items_d9f67aa6, ix_stock_return_items_7032963c, ix_stock_return_items_10e3c1ce, ix_stock_return_items_876d7bdb, stock_return_items_pos_sale_item_id_foreign
suppliers	0	PRIMARY, uq_suppliers_e1183bab, ix_suppliers_6ae99955, ix_suppliers_e7e1e7cc
supplier_account_entries	0	PRIMARY, uq_supplier_account_entries_4e0db050, uq_supplier_account_entries_7db4ecce, uq_supplier_account_entries_1de3c56e, uq_supplier_account_entries_86cc2ebe, uq_supplier_account_entries_69c70222, uq_supplier_account_entries_3ec7caab, uq_supplier_account_entries_fc740855, fk_supplier_account_entries_10afd45a, fk_supplier_account_entries_3e1bac6a, fk_supplier_account_entries_e24bcc51, fk_supplier_account_entries_80141e8c, ix_supplier_account_entries_a3fe3b0b, ix_supplier_account_entries_2bc0d2ca, ix_supplier_account_entries_06129e94
supplier_allocation_releases	0	PRIMARY, uq_supplier_allocation_releases_7db4ecce, fk_supplier_allocation_releases_fd2690eb, fk_supplier_allocation_releases_2e689253, ix_supplier_allocation_releases_ba299898, ix_supplier_allocation_releases_69c70222, ix_supplier_allocation_releases_3ec7caab, ix_supplier_allocation_releases_e78980db
supplier_payments	0	PRIMARY, uq_supplier_payments_4e0db050, uq_supplier_payments_39743c6d, uq_supplier_payments_7db4ecce, ix_supplier_payments_7c3ebe1b, ix_supplier_payments_2bc0d2ca, ix_supplier_payments_38e2e5db

Table	Rows at read-only snapshot	Index names
supplier_payment_allocations	0	PRIMARY, uq_supplier_payment_allocations_3f75acb8, uq_supplier_payment_allocations_7db4ecce, fk_supplier_payment_allocations_e24bcc51, fk_supplier_payment_allocations_3e1bac6a, ix_supplier_payment_allocations_69c70222, ix_supplier_payment_allocations_2bc0d2ca, ix_supplier_payment_allocations_86cc2ebe, ix_supplier_payment_allocations_997c1d72
supplier_payment_reversals	0	PRIMARY, uq_supplier_payment_reversals_3f75acb8, uq_supplier_payment_reversals_4e0db050, uq_supplier_payment_reversals_7db4ecce, fk_supplier_payment_reversals_e24bcc51, ix_supplier_payment_reversals_69c70222, ix_supplier_payment_reversals_2bc0d2ca, ix_supplier_payment_reversals_07fdd642
system_settings	0	PRIMARY, uq_system_settings_3b49c75a, ix_system_settings_7a022eb2
units_of_measure	0	PRIMARY, uq_units_of_measure_e6fb0621
users	7	PRIMARY, users_email_unique, uq_users_username

Declared foreign-key mappings

Child	Parent	Delete rule	Orphans
audit_logs.user_id	users.id	RESTRICT	0
backup_histories.initiated_by_user_id	users.id	RESTRICT	0

Child	Parent	Delete rule	Orphans
backup_restores.initiated_by_user_id	users.id	RESTRICT	0
backup_restores.backup_history_id	backup_histories.id	RESTRICT	0
customer_account_entries.invoice_id	invoices.id	RESTRICT	0
customer_account_entries.reverses_entry_id	customer_account_entries.id	RESTRICT	0
customer_account_entries.payment_id	payments.id	RESTRICT	0
customer_account_entries.pos_sale_id	pos_sales.id	RESTRICT	0
customer_account_entries.pos_sale_id	pos_sales.id	RESTRICT	0
customer_account_entries.customer_id	pos_sales.customer_id	RESTRICT	0
customer_account_entries.payment_reversal_id	payment_reversals.id	RESTRICT	0
customer_account_entries.customer_id	payment_reversals.customer_id	RESTRICT	0
customer_account_entries.payment_id	payments.id	RESTRICT	0
customer_account_entries.customer_id	payments.customer_id	RESTRICT	0
customer_account_entries.invoice_id	invoices.id	RESTRICT	0
customer_account_entries.customer_id	invoices.customer_id	RESTRICT	0
customer_account_entries.payment_reversal_id	payment_reversals.id	RESTRICT	0
customer_account_entries.recorded_by_user_id	users.id	RESTRICT	0
customer_account_entries.reverses_entry_id	customer_account_entries.id	RESTRICT	0
customer_account_entries.customer_id	customer_account_entries.customer_id	RESTRICT	0

Child	Parent	Delete rule	Orphans
customer_account_entries.customer_id	customers.id	RESTRICT	0
document_print_histories.invoice_id	invoices.id	RESTRICT	0
document_print_histories.repair_receipt_id	repair_receipts.id	RESTRICT	0
document_print_histories.payment_id	payments.id	RESTRICT	0
document_print_histories.printed_by_user_id	users.id	RESTRICT	0
document_print_histories.pos_sale_id	pos_sales.id	RESTRICT	0
document_print_histories.purchase_id	purchases.id	RESTRICT	0
document_print_histories.quotation_id	quotations.id	RESTRICT	0
document_print_histories.job_card_id	job_cards.id	RESTRICT	0
document_print_histories.print_template_id	print_templates.id	RESTRICT	0
document_print_histories.customer_id	customers.id	RESTRICT	0
employees.user_id	users.id	RESTRICT	0
expenses.expense_category_id	expense_categories.id	RESTRICT	0
expenses.recorded_by_user_id	users.id	RESTRICT	0
goods_received_notes.purchase_id	purchases.id	RESTRICT	0
goods_received_notes.supplier_id	purchases.supplier_id	RESTRICT	0
goods_received_notes.purchase_id	purchases.id	RESTRICT	0
goods_received_notes.received_by_employee_id	employees.id	RESTRICT	0
goods_received_notes.supplier_id	suppliers.id	RESTRICT	0

Child	Parent	Delete rule	Orphans
goods_received_note_items.purchase_item_id	purchase_items.id	RESTRICT	0
goods_received_note_items.purchase_item_id	purchase_items.id	RESTRICT	0
goods_received_note_items.spare_part_id	purchase_items.spare_part_id	RESTRICT	0
goods_received_note_items.spare_part_id	spare_parts.id	RESTRICT	0
goods_received_note_items.location_id	stock_locations.id	RESTRICT	0
goods_received_note_items.purchase_id	purchases.id	RESTRICT	0
goods_received_note_items.goods_received_note_id	goods_received_notes.id	RESTRICT	0
goods_received_note_items.purchase_item_id	purchase_items.id	RESTRICT	0
goods_received_note_items.purchase_id	purchase_items.purchase_id	RESTRICT	0
goods_received_note_items.goods_received_note_id	goods_received_notes.id	RESTRICT	0
goods_received_note_items.purchase_id	goods_received_notes.purchase_id	RESTRICT	0
invoices.cashier_employee_id	employees.id	RESTRICT	0
invoices.cancelled_by_user_id	users.id	RESTRICT	0
invoices.customer_id	customers.id	RESTRICT	0
invoice_history.actor_user_id	users.id	RESTRICT	0
invoice_history.invoice_id	invoices.id	RESTRICT	0
invoice_items.invoice_id	invoices.id	RESTRICT	0
invoice_items.quotation_item_id	quotation_items.id	RESTRICT	0
invoice_items.spare_part_id	spare_parts.id	RESTRICT	0
invoice_items.invoice_job_id	invoice_jobs.id	RESTRICT	0
invoice_items.invoice_job_id	invoice_jobs.id	RESTRICT	0
invoice_items.invoice_id	invoice_jobs.invoice_id	RESTRICT	0

Child	Parent	Delete rule	Orphans
invoice_jobs.repair_job_id	repair_jobs.id	RESTRICT	0
invoice_jobs.customer_id	repair_jobs.customer_id	RESTRICT	0
invoice_jobs.invoice_id	invoices.id	RESTRICT	0
invoice_jobs.repair_job_id	repair_jobs.id	RESTRICT	0
invoice_jobs.invoice_id	invoices.id	RESTRICT	0
invoice_jobs.customer_id	invoices.customer_id	RESTRICT	0
invoice_jobs.customer_id	customers.id	RESTRICT	0
job_approvals.repair_job_id	repair_jobs.id	RESTRICT	0
job_approvals.recorded_by_user_id	users.id	RESTRICT	0
job_cards.repair_job_id	repair_jobs.id	RESTRICT	0
job_cards.print_template_id	print_templates.id	RESTRICT	0
job_inspections.inspected_by_employee_id	employees.id	RESTRICT	0
job_inspections.repair_job_id	repair_jobs.id	RESTRICT	0
job_inspections.repaired_by_employee_id	employees.id	RESTRICT	0
job_part_usages.spare_part_id	spare_parts.id	RESTRICT	0
job_part_usages.repair_job_id	repair_jobs.id	RESTRICT	0
job_part_usages.approved_by_user_id	users.id	RESTRICT	0
job_part_usages.location_id	stock_locations.id	RESTRICT	0
job_part_usages.requested_by_user_id	users.id	RESTRICT	0
job_status_histories.repair_job_id	repair_jobs.id	RESTRICT	0
job_status_histories.changed_by_user_id	users.id	RESTRICT	0
payments.received_by_employee_id	employees.id	RESTRICT	0
payments.customer_id	customers.id	RESTRICT	0
payment_allocations.repair_job_id	repair_jobs.id	RESTRICT	0

Child	Parent	Delete rule	Orphans
payment_allocations.customer_id	repair_jobs.customer_id	RESTRICT	0
payment_allocations.invoice_id	invoices.id	RESTRICT	0
payment_allocations.payment_id	payments.id	RESTRICT	0
payment_allocations.pos_sale_id	pos_sales.id	RESTRICT	0
payment_allocations.pos_sale_id	pos_sales.id	RESTRICT	0
payment_allocations.customer_id	pos_sales.customer_id	RESTRICT	0
payment_allocations.payment_id	payments.id	RESTRICT	0
payment_allocations.customer_id	payments.customer_id	RESTRICT	0
payment_allocations.payment_id	payments.id	RESTRICT	0
payment_allocations.customer_scope_id	payments.customer_scope_id	RESTRICT	0
payment_allocations.repair_job_id	repair_jobs.id	RESTRICT	0
payment_allocations.invoice_id	invoices.id	RESTRICT	0
payment_allocations.customer_id	invoices.customer_id	RESTRICT	0
payment_allocations.allocated_by_user_id	users.id	RESTRICT	0
payment_allocations.pos_sale_id	pos_sales.id	RESTRICT	0
payment_allocations.customer_scope_id	pos_sales.customer_scope_id	RESTRICT	0
payment_allocations.customer_id	customers.id	RESTRICT	0
payment_allocation_releases.payment_id	payments.id	RESTRICT	0

Child	Parent	Delete rule	Orphans
payment_allocation_releases.payment_allocation_id	payment_allocations.id	RESTRICT	0
payment_allocation_releases.released_by_user_id	users.id	RESTRICT	0
payment_allocation_releases.payment_reversal_id	payment_reversals.id	RESTRICT	0
payment_allocation_releases.payment_reversal_id	payment_reversals.id	RESTRICT	0
payment_allocation_releases.payment_id	payment_reversals.payment_id	RESTRICT	0
payment_allocation_releases.payment_allocation_id	payment_allocations.id	RESTRICT	0
payment_allocation_releases.payment_id	payment_allocations.payment_id	RESTRICT	0
payment_reversals.payment_id	payments.id	RESTRICT	0
payment_reversals.reversed_by_user_id	users.id	RESTRICT	0
payment_reversals.payment_id	payments.id	RESTRICT	0
payment_reversals.customer_scope_id	payments.customer_scope_id	RESTRICT	0
payment_reversals.customer_id	customers.id	RESTRICT	0
permission_role.role_id	roles.id	RESTRICT	0
permission_role.permission_id	permissions.id	RESTRICT	0
permission_user.user_id	users.id	RESTRICT	0
permission_user.permission_id	permissions.id	RESTRICT	0
pos_sales.cashier_employee_id	employees.id	RESTRICT	0
pos_sales.customer_id	customers.id	RESTRICT	0
pos_sales.payment_verified_by_user_id	users.id	RESTRICT	0
pos_sale_items.pos_sale_id	pos_sales.id	RESTRICT	0
pos_sale_items.spare_part_id	spare_parts.id	RESTRICT	0

Child	Parent	Delete rule	Orphans
pos_sale_items.location_id	stock_locations.id	RESTRICT	0
purchases.created_by_user_id	users.id	RESTRICT	0
purchases.supplier_id	suppliers.id	RESTRICT	0
purchase_items.spare_part_id	spare_parts.id	RESTRICT	0
purchase_items.purchase_id	purchases.id	RESTRICT	0
quotations.repair_job_id	repair_jobs.id	RESTRICT	0
quotations.customer_id	repair_jobs.customer_id	RESTRICT	0
quotations.created_by_user_id	users.id	RESTRICT	0
quotations.repair_job_id	repair_jobs.id	RESTRICT	0
quotations.revises_quotation_id	quotations.id	RESTRICT	0
quotations.revises_quotation_id	quotations.id	RESTRICT	0
quotations.repair_job_id	quotations.repair_job_id	RESTRICT	0
quotations.customer_id	quotations.customer_id	RESTRICT	0
quotations.customer_id	customers.id	RESTRICT	0
quotations.updated_by_user_id	users.id	RESTRICT	0
quotation_approvals.recorded_by_user_id	users.id	RESTRICT	0
quotation_approvals.quotation_id	quotations.id	RESTRICT	0
quotation_history.actor_user_id	users.id	RESTRICT	0
quotation_history.quotation_id	quotations.id	RESTRICT	0
quotation_items.spare_part_id	spare_parts.id	RESTRICT	0
quotation_items.quotation_id	quotations.id	RESTRICT	0
repair_jobs.technician_id	employees.id	RESTRICT	0
repair_jobs.received_by_employee_id	employees.id	RESTRICT	0
repair_jobs.cancelled_by_user_id	users.id	RESTRICT	0
repair_jobs.customer_id	customers.id	RESTRICT	0

Child	Parent	Delete rule	Orphans
repair_receipts.repair_job_id	repair_jobs.id	RESTRICT	0
repair_receipts.customer_id	repair_jobs.customer_id	RESTRICT	0
repair_receipts.repair_job_id	repair_jobs.id	RESTRICT	0
repair_receipts.customer_id	customers.id	RESTRICT	0
role_user.role_id	roles.id	RESTRICT	0
role_user.user_id	users.id	RESTRICT	0
spare_parts.default_location_id	stock_locations.id	RESTRICT	0
spare_parts.unit_id	units_of_measure.id	RESTRICT	0
spare_parts.category_id	spare_part_categories.id	RESTRICT	0
spare_parts.supplier_id	suppliers.id	RESTRICT	0
spare_parts.brand_id	spare_part_brands.id	RESTRICT	0
stock_adjustments.created_by_user_id	users.id	RESTRICT	0
stock_adjustments.approved_by_user_id	users.id	RESTRICT	0
stock_adjustment_items.stock_adjustment_id	stock_adjustments.id	RESTRICT	0
stock_adjustment_items.spare_part_id	spare_parts.id	RESTRICT	0
stock_adjustment_items.location_id	stock_locations.id	RESTRICT	0
stock_counts.counted_by_user_id	users.id	RESTRICT	0
stock_count_items.stock_count_id	stock_counts.id	RESTRICT	0
stock_count_items.spare_part_id	spare_parts.id	RESTRICT	0
stock_count_items.location_id	stock_locations.id	RESTRICT	0
stock_movements.job_part_usage_id	job_part_usages.id	RESTRICT	0
stock_movements.spare_part_id	job_part_usages.spare_part_id	RESTRICT	0
stock_movements.location_id	job_part_usages.location_id	RESTRICT	0

Child	Parent	Delete rule	Orphans
stock_movements.count_item_id	stock_count_items.id	RESTRICT	0
stock_movements.adjustment_item_id	stock_adjustment_items.id	RESTRICT	0
stock_movements.spare_part_id	stock_adjustment_items.spare_part_id	RESTRICT	0
stock_movements.location_id	stock_adjustment_items.location_id	RESTRICT	0
stock_movements.reverses_movement_id	stock_movements.id	RESTRICT	0
stock_movements.spare_part_id	spare_parts.id	RESTRICT	0
stock_movements.reverses_movement_id	stock_movements.id	RESTRICT	0
stock_movements.spare_part_id	stock_movements.spare_part_id	RESTRICT	0
stock_movements.location_id	stock_movements.location_id	RESTRICT	0
stock_movements.location_id	stock_locations.id	RESTRICT	0
stock_movements.recorded_by_user_id	users.id	RESTRICT	0
stock_movements.adjustment_item_id	stock_adjustment_items.id	RESTRICT	0
stock_movements.return_item_id	stock_return_items.id	RESTRICT	0
stock_movements.spare_part_id	stock_return_items.spare_part_id	RESTRICT	0
stock_movements.location_id	stock_return_items.location_id	RESTRICT	0
stock_movements.return_item_id	stock_return_items.id	RESTRICT	0
stock_movements.pos_sale_item_id	pos_sale_items.id	RESTRICT	0
stock_movements.pos_sale_item_id	pos_sale_items.id	RESTRICT	0
stock_movements.spare_part_id	pos_sale_items.spare_part_id	RESTRICT	0
stock_movements.location_id	pos_sale_items.location_id	RESTRICT	0

Child	Parent	Delete rule	Orphans
stock_movements.job_part_usage_id	job_part_usages.id	RESTRICT	0
stock_movements.grn_item_id	goods_received_note_items.id	RESTRICT	0
stock_movements.spare_part_id	goods_received_note_items.spare_part_id	RESTRICT	0
stock_movements.location_id	goods_received_note_items.location_id	RESTRICT	0
stock_movements.grn_item_id	goods_received_note_items.id	RESTRICT	0
stock_movements.count_item_id	stock_count_items.id	RESTRICT	0
stock_movements.spare_part_id	stock_count_items.spare_part_id	RESTRICT	0
stock_movements.location_id	stock_count_items.location_id	RESTRICT	0
stock_returns.pos_sale_id	pos_sales.id	RESTRICT	0
stock_returns.repair_job_id	repair_jobs.id	RESTRICT	0
stock_returns.purchase_id	purchases.id	RESTRICT	0
stock_returns.recorded_by_user_id	users.id	RESTRICT	0
stock_returns.refund_payment_reversal_id	payment_reversals.id	RESTRICT	0
stock_return_items.spare_part_id	spare_parts.id	RESTRICT	0
stock_return_items.location_id	stock_locations.id	RESTRICT	0
stock_return_items.stock_return_id	stock_returns.id	RESTRICT	0
stock_return_items.pos_sale_item_id	pos_sale_items.id	RESTRICT	0
supplier_account_entries.supplier_payment_reversal_id	supplier_payment_reversals.id	RESTRICT	0
supplier_account_entries.supplier_id	supplier_payment_reversals.supplier_id	RESTRICT	0
supplier_account_entries.purchase_id	purchases.id	RESTRICT	0
supplier_account_entries.supplier_id	purchases.supplier_id	RESTRICT	0

Child	Parent	Delete rule	Orphans
supplier_account_entries.reverses_entry_id	supplier_account_entries.id	RESTRICT	0
supplier_account_entries.reverses_entry_id	supplier_account_entries.id	RESTRICT	0
supplier_account_entries.supplier_id	supplier_account_entries.supplier_id	RESTRICT	0
supplier_account_entries.purchase_id	purchases.id	RESTRICT	0
supplier_account_entries.supplier_payment_id	supplier_payments.id	RESTRICT	0
supplier_account_entries.supplier_payment_reversal_id	supplier_payment_reversals.id	RESTRICT	0
supplier_account_entries.recorded_by_user_id	users.id	RESTRICT	0
supplier_account_entries.supplier_id	suppliers.id	RESTRICT	0
supplier_account_entries.supplier_payment_id	supplier_payments.id	RESTRICT	0
supplier_account_entries.supplier_id	supplier_payments.supplier_id	RESTRICT	0
supplier_allocation_releases.supplier_payment_reversal_id	supplier_payment_reversals.id	RESTRICT	0
supplier_allocation_releases.supplier_payment_id	supplier_payment_reversals.supplier_payment_id	RESTRICT	0
supplier_allocation_releases.released_by_user_id	users.id	RESTRICT	0
supplier_allocation_releases.supplier_payment_allocation_id	supplier_payment_allocations.id	RESTRICT	0
supplier_allocation_releases.supplier_payment_id	supplier_payments.id	RESTRICT	0
supplier_allocation_releases.supplier_payment_reversal_id	supplier_payment_reversals.id	RESTRICT	0
supplier_allocation_releases.supplier_payment_allocation_id	supplier_payment_allocations.id	RESTRICT	0
supplier_allocation_releases.supplier_payment_id	supplier_payment_allocations.supplier_payment_id	RESTRICT	0

Child	Parent	Delete rule	Orphans
supplier_payments.paid_by_user_id	users.id	RESTRICT	0
supplier_payments.supplier_id	suppliers.id	RESTRICT	0
supplier_payment_allocations.purchase_id	purchases.id	RESTRICT	0
supplier_payment_allocations.supplier_id	purchases.supplier_id	RESTRICT	0
supplier_payment_allocations.purchase_id	purchases.id	RESTRICT	0
supplier_payment_allocations.supplier_payment_id	supplier_payments.id	RESTRICT	0
supplier_payment_allocations.allocated_by_user_id	users.id	RESTRICT	0
supplier_payment_allocations.supplier_id	suppliers.id	RESTRICT	0
supplier_payment_allocations.supplier_payment_id	supplier_payments.id	RESTRICT	0
supplier_payment_allocations.supplier_id	supplier_payments.supplier_id	RESTRICT	0
supplier_payment_reversals.reversed_by_user_id	users.id	RESTRICT	0
supplier_payment_reversals.supplier_payment_id	supplier_payments.id	RESTRICT	0
supplier_payment_reversals.supplier_id	suppliers.id	RESTRICT	0
supplier_payment_reversals.supplier_payment_id	supplier_payments.id	RESTRICT	0
supplier_payment_reversals.supplier_id	supplier_payments.supplier_id	RESTRICT	0
system_settings.updated_by_user_id	users.id	RESTRICT	0

Follow-up exact tuple audit inspected 199 complete foreign-key constraints, including composite keys; orphan tuples: 0. Before/after operational table row counts are identical (`step12-preservation.log`). Row-count equality is not a byte-for-byte data comparison.

Financial, stock and history records use restrictive delete behavior tested in SchemaIntegrityTest; authentication/permission infrastructure has its own cascade rules. Polymorphic audit identities intentionally have no target FK. Models/relationships are tested separately; supplier/purchase relationships do not establish

operational CRUD support.

Pending-work follow-up

No schema changes were necessary. All workflows reuse the existing 69 tables and declared keys. Follow-up before/after snapshots and exact composite-key orphan checks are recorded separately. See pending-preservation.log; row-count equality is not a byte-for-byte data comparison.

Security test report

Safe local testing only. AuthenticationTest executes password rejection, CSRF enforcement, five-attempt rate limiting, session regeneration, logout invalidation, disabled-account rejection and no public registration. EmployeeManagementTest tests all six role definitions, per-user denial precedence, direct route restrictions, assigned technician access and last-administrator protection. Payment/POS/backup tests exercise sensitive operation permissions.

Step12AuditTest sends an SQL-looking search value and verifies it remains data, then verifies stored script markup is escaped in customer details. Report sort fields are allowlisted. These narrow probes do not constitute exhaustive penetration testing of every field.

SystemSupportTest rejects unknown/secret editable settings; AuditSafetyTest covers redaction and historical log safety. Backup tests reject web-root destinations and traversal, conceal private paths, verify checksums and require authenticated restore confirmation. Google Drive transport is mocked; no actual account access is configured.

PHP syntax and Composer validation results are in their evidence logs. Composer audit reported no security vulnerability advisories at execution time; this is a dependency advisory result, not proof the application has no vulnerabilities. No package versions were changed.

Unverified: full hostile upload corpus, every stored/reflected XSS sink, end-to-end browser session timeout, local filesystem permission guarantees for every Windows account, production HTTPS/cookie configuration (outside this phase). No security mechanism was disabled to pass business tests. Supplier/stock authorization cannot be accepted for absent workflows. The conflicting management/storekeeper purchase rule requires owner agreement.

Pending-work follow-up

New routes inherit auth/active-account/CSRF middleware. Services also enforce permissions for financial posting, stock operations and assigned-job requests. Cashiers cannot administer suppliers/stock; storekeepers cannot post supplier payments or approve purchases using the existing role matrix. Quoted, billed and closed job charges cannot be changed through the unquoted pricing endpoint. No credential values were introduced into settings or reports. New SQL filters are parameterized; print fields remain escaped. The prior text describing supplier/stock operations as absent is superseded by this section and the final workflow suite.

Financial reconciliation report

Final native MySQL result: **218 passed (3566 assertions)**, duration 175.11s. Full evidence: `storage/logs/pending-complete-mysql.log`. The helper created its own randomized database, ran tests, rolled back/remigrated that disposable database, ran native multiuser/recovery checks, then removed only that database. Operational tables were not migrated/reset.

Focused evidence: `pending-focused-final.log`; PDF dimensions/fonts: `pending-pdf-review.json`; before/after operational schema/count evidence: `pending-schema-before.json`, `pending-schema-after.json`, `pending-preservation.log`; syntax/style evidence: `pending-syntax.log`, `pending-style-check.log`. All names are under `storage/logs/`.

Exact disposable-fixture assertions include: approved quotation total 30.00, advance 5.00, outstanding 25.00 then 0.00; repair parts cost 6.00 and labour cost 2.00 produce recorded profit estimate 22.00. Supplier purchase total 16.00, payment 6.00 leaves 10.00; full reversal restores 16.00. Agreed supplier return credit 5.00 leaves 11.00. Opening balance 10.50 less payment 3.50 gives 7.00, restored to 10.50 after reversal. These are test fixtures only.

Latest-purchase fixture: an old sale costs 3.00, a new purchase costs 7.00, the old item returns at its original 3.00, and a subsequent sale still snapshots 7.00. Nine units are valued at 63.00. Net revenue 10.00 and net recovered/sold cost 7.00 produce a 3.00 recorded margin. Unknown costs remain unknown, not invented zero costs.

Invoice and customer payment calculations continue to derive net allocations/releases; advances are not counted twice. Supplier balances use the existing supplier ledger, with positive purchase/opening liabilities, negative payments/return credits, and linked positive reversals. Original entries remain append-only. No parallel accounting ledger was created.

Latest purchase cost is owner-approved. Purchase-level freight/discount allocation is not yet approved. Affected estimates, cost mismatches, unknown source costs, no-refund stock returns and ambiguous repair cash refunds remain labelled incomplete. Profit is not sales revenue or a certified accounting statement.

Inventory reconciliation report

Final native MySQL result: **218 passed (3566 assertions)**, duration 175.11s. Full evidence: `storage/logs/pending-complete-mysql.log`. The helper created its own randomized database, ran tests, rolled back/remigrated that disposable database, ran native multiuser/recovery checks, then removed only that database. Operational tables were not migrated/reset.

Focused evidence: `pending-focused-final.log`; PDF dimensions/fonts: `pending-pdf-review.json`; before/after operational schema/count evidence: `pending-schema-before.json`, `pending-schema-after.json`, `pending-preservation.log`; syntax/style evidence: `pending-syntax.log`, `pending-style-check.log`. All names are under `storage/logs/`.

One signed movement ledger supports opening stock, GRNs, authorized repair issues, POS, resalable returns, supplier returns, damage, corrections and physical counts. The part is locked before checking current ledger quantities, using the same product lock as POS. Requests do not reserve or deduct stock; approval-and-issue records both authorization and movement atomically. Complete unused issues may be returned once before billing, retaining the original negative movement and a linked positive return.

Native shared-stock race: repair and POS each request four from five; one succeeds and one rejects, leaving one unit. Native GRN race: two receivers request four against five ordered; exactly four are received once. Sequential replay, over-receiving, counts and damage cases also pass. Negative stock is rejected by the workflow. Counts record expected and observed quantities before posting their difference.

Valuation uses the latest posted purchase cost at the location, falling back to an explicitly known opening cost only when no purchase cost exists. Old returns retain original cost snapshots and do not alter the selected latest purchase cost. Unknown costs are visibly excluded from known valuation totals. Purchase-wide adjustment allocation remains pending.

Local performance test report

Final native MySQL result: **218 passed (3566 assertions)**, duration 175.11s. Full evidence: `storage/logs/pending-complete-mysql.log`. The helper created its own randomized database, ran tests, rolled back/remigrated that disposable database, ran native multiuser/recovery checks, then removed only that database. Operational tables were not migrated/reset.

Focused evidence: `pending-focused-final.log`; PDF dimensions/fonts: `pending-pdf-review.json`; before/after operational schema/count evidence: `pending-schema-before.json`, `pending-schema-after.json`, `pending-preservation.log`; syntax/style evidence: `pending-syntax.log`, `pending-style-check.log`. All names are under `storage/logs/`.

One local service iteration per operation; fixture construction excluded; no browser/network latency.

Operation	ms
invoice-create-and-issue	57.288
payment-receive-and-allocate	63.399
pos-checkout-including-stock-posting	34.648
a5-invoice-pdf-with-print-history	1424.018

Dataset row counts: `{"customers": 2, "repair_jobs": 2, "invoices": 2, "payments": 7, "pos_sales": 5, "spare_parts": 3, "stock_movements": 8}`. Native race timings are in the completion report. These are small local observations, not percentiles or production capacity. Existing report/dashboard/customer/job timing fixture output remains in step12-local-timings.json. No realistic-volume capacity or statistically defensible slow-query benchmark was performed.

Printing and PDF verification

Final native MySQL result: **218 passed (3566 assertions)**, duration 175.11s. Full evidence: `storage/logs/pending-complete-mysql.log`. The helper created its own randomized database, ran tests, rolled back/remigrated that disposable database, ran native multiuser/recovery checks, then removed only that database. Operational tables were not migrated/reset.

Focused evidence: `pending-focused-final.log`; PDF dimensions/fonts: `pending-pdf-review.json`; before/after operational schema/count evidence: `pending-schema-before.json`, `pending-schema-after.json`, `pending-preservation.log`; syntax/style evidence: `pending-syntax.log`, `pending-style-check.log`. All names are under `storage/logs/`.

New fixture PDFs were generated, rendered to PNG and visually inspected with embedded regular/bold Noto Sinhala fonts: A5 repair receipt (one page), A6 job card (two pages), customized A5 landscape statement (one page), A4 supplier purchase and GRN (one page each). Exact dimensions/font metadata are in pending-pdf-review.json; artifacts are under storage/app/testing/pending-*.pdf and *-review-*.png. Earlier overflow and missing-glyph defects were corrected. Long documents may legitimately paginate beyond these short-fixture counts.

Statement paper/orientation/title now drive both browser CSS and PDF paper, and statement prints/duplicates preserve history. Invoice visible heading uses its configured title. POS receipt PDFs now embed Sinhala fonts and honor configured font size, including thermal output. Intake PDF download/reprint routes enforce original permissions and retain duplicate labels. Supplier/GRN documents support browser print, PDF, duplication, branding, margins, signatures and audit history using the existing purchase print template.

No physical printer/scanner or fresh interactive browser print-preview acceptance was performed. Physical margins, two-sided card handling, thermal roll behavior and native-language business approval remain owner/hardware UAT items. No asset build or packaging occurred.

Backup and restore test report

Schedule definition: daily `0 2 * * *`, Asia/Colombo; configured retention 30 days, maximum 30 scheduled daily copies. Manual copies are protected. `step12-schedule.log` records schedule inspection. No production/Windows scheduler was installed or enabled.

Step11BackupTest verifies retention, latest recoverable copy protection, history retention, changed/missing/path-traversal detection, private destination checks, blank Drive disabling and mocked remote checksum/size verification. BackupManagementTest tests authorization, history and failures.

Native helper actually creates a nonempty MySQL dump, verifies SHA-256, rejects absent confirmation/wrong password, creates a safety dump, restores altered fixture data, checks recovered data/audit/history and maintenance cleanup. It runs only against the newly created disposable MySQL database and an isolated temporary storage path. Evidence: `PASS: real MySQL dump...` and `PASS: confirmation and wrong-password refusals...` in full-run logs. This is not a restore over the operational database.

Operational destination remains blank pending an explicitly approved location. Earlier automatic approval review rejected exporting live data to an unspecified/unapproved backup destination; no such export was performed in Step 12. Google Drive folder/account/OAuth credentials remain blank as requested. Real cloud upload, operational scheduled execution and off-computer recovery are therefore blocked, not passed. No credential values are reproduced here.

Pending-work follow-up

Real native dump/restore and confirmation/password refusal checks were repeated successfully in pending-complete-mysql.log after the new modules. Operational destination/Drive/scheduler status remains as above.

Defect register

Final evidence: 218 passed (3566 assertions); see pending-complete-mysql.log. Earlier failures remain in their original evidence logs and were not hidden.

ID	Severity	Finding	Current state
D12-01	Medium	Discounted refund cent rounding	Fixed and regression-tested.
D12-02	Low	MySQL invoice rollback fault-injection coverage	Portable model-event injection passes.
G12-01	High	Missing inventory issue/receiving/adjustment/count operations	Implemented and tested; full unused-issue return supported.
G12-02	High	Missing suppliers/purchases/GRNs/settlement	Implemented and tested, including opening settlement and return credits.
D12-03	High	Approved quotation did not transfer job selling totals	Fixed; receipt-to-paid-invoice test uses real service calls, without manually setting job prices.
D12-04	Medium	Completion/delivery statuses could leave timestamps empty	Fixed; timestamp assertions pass.
D12-05	Medium	Older returned stock could replace newer purchase cost	Fixed with owner-approved latest-purchase source selection.
D12-06	Medium	Accounting payment reversal reduced reported sales	Fixed; collections change while issued sale remains intact.
G12-04	Medium	Intake PDF gaps, fonts, statement title/paper propagation	Implemented and tested; short fixture PDFs visually reviewed. Physical acceptance pending.
G12-03	High	Operational backup destination/cloud account unset	Awaiting approved private folder; Drive remains blank by request. No host scheduler activated.
G12-05	Medium	Purchase-wide cost allocation / ambiguous financial treatment	Latest purchase method approved; freight/discount allocation still awaits answer. Affected profit stays incomplete.

ID	Severity	Finding	Current state
G12-06	Medium	Reception/technician/repair-POS/GRN concurrency	Two-worker native cases pass. Realistic-volume capacity remains unmeasured.
G12-07	Medium	Physical printing/scanning and business sign-off	External UAT pending; no signatures fabricated.
L12-01	Low	Partial return of one repair issue / repeated-part historical purchase return	Full unused repair issue can be reversed; ambiguous repeated-part purchase return is refused pending source reconciliation.

User acceptance test cases

Tester for automated evidence: Codex local test runner, 2026-10-20. Business tester/signature: pending. Automation does not substitute for business acceptance. Execute manual cases only in an approved isolated/UAT database.

UAT-01: Register individual and company

- BRD reference: 4.
- Preconditions: Receptionist with customer permission.
- Steps: Enter required contacts; save both types; search by ID/name/mobile.
- Expected: Unique IDs, validation and matching search results.
- Actual: Automated customer suite; manual sign-off pending.
- Status: Blocked for business acceptance.
- Tester: automated runner where cited; business tester pending.
- Date: 2026-10-20 audit; manual date pending.
- Evidence: `CustomerManagementTest`; final execution logs.

UAT-02: Atomic receipt and job/card

- BRD reference: 5–6.
- Preconditions: Registered customer and intake role.
- Steps: Enter manual number, appliance, advance; save; repeat number; print A5/A6.
- Expected: One linked job/receipt, no duplicate advance; correct duplicate label.
- Actual: Automated intake/advance assertions; physical output pending.
- Status: Blocked for business acceptance.
- Tester: automated runner where cited; business tester pending.
- Date: 2026-10-20 audit; manual date pending.
- Evidence: `RepairReceiptAndJobManagementTest; PaymentManagementTest`; final execution logs.

UAT-03: Inspection and quotation approval

- BRD reference: 6–8.

- Preconditions: Assigned technician and authorized decision recorder.
- Steps: Record inspection; draft/issue quote; approve; revise another quote.
- Expected: History appended; exact prices; no stock or payment created by approval.
- Actual: Automated quotation/inspection suite; manual acceptance pending.
- Status: Blocked for business acceptance.
- Tester: automated runner where cited; business tester pending.
- Date: 2026-10-20 audit; manual date pending.
- Evidence: `QuotationManagementTest`; final execution logs.

UAT-04: Repair with approved part issue

- BRD reference: 12,20.
- Preconditions: Stock available; approved quotation.
- Steps: Approve part request; issue once; repeat request; complete job.
- Expected: Only issued approved quantity deducted once; auditable cost.
- Actual: Issue workflow missing.
- Status: Blocked for business acceptance.
- Tester: automated runner where cited; business tester pending.
- Date: 2026-10-20 audit; manual date pending.
- Evidence: `G12-01`; final execution logs.

UAT-05: Multi-job invoice and payment

- BRD reference: 9–11.
- Preconditions: Two completed same-customer jobs; one advance.
- Steps: Issue invoice; partial cash/bank payment; collect balance; reverse authorized payment.
- Expected: No double billing/advance; total minus net payments; immutable original.
- Actual: Automated invoice/payment tests; manual acceptance pending.
- Status: Blocked for business acceptance.
- Tester: automated runner where cited; business tester pending.
- Date: 2026-10-20 audit; manual date pending.

- Evidence: `InvoiceManagementTest; PaymentManagementTest`; final execution logs.

UAT-06: Delivery and reports

- BRD reference: 6,20.
- Preconditions: Paid repair ready for collection.
- Steps: Record collector/vehicle/date; deliver; filter reports and export.
- Expected: Correct delivery status/history and totals.
- Actual: Status/report tests exist; full workflow blocked by stock issue.
- Status: Blocked for business acceptance.
- Tester: automated runner where cited; business tester pending.
- Date: 2026-10-20 audit; manual date pending.
- Evidence: `RepairReceiptAndJobManagementTest; Step11ReportingTest`; final execution logs.

UAT-07: POS sale and return

- BRD reference: 12,14,23.
- Preconditions: Cashier; manager; isolated stock.
- Steps: Scan; sell; retry checkout; return resalable/damaged parts; refund discounted sale.
- Expected: One sale/payment/movement; authorized exact refund; original retained.
- Actual: Automated POS and two-worker races; physical scanner/manual approval pending.
- Status: Blocked for business acceptance.
- Tester: automated runner where cited; business tester pending.
- Date: 2026-10-20 audit; manual date pending.
- Evidence: `POSManagementTest; verify-pos-concurrency.php`; final execution logs.

UAT-08: Supplier purchase and partial receiving

- BRD reference: 13.
- Preconditions: Storekeeper/management scope agreed.
- Steps: Create supplier and draft purchase; partial GRN twice; pay partially; reverse payment.
- Expected: Draft no stock; each GRN posts once; correct supplier balance.

- Actual: Supplier/GRN/settlement services missing.
- Status: Blocked for business acceptance.
- Tester: automated runner where cited; business tester pending.
- Date: 2026-10-20 audit; manual date pending.
- Evidence: `G12-02`; final execution logs.

UAT-09: Backup and disposable recovery

- BRD reference: 17.
- Preconditions: Approved destination and administrator.
- Steps: Create dump; verify hash; alter disposable probe; confirm/authenticate restore.
- Expected: Probe recovered, safety copy and audit present; no live DB change.
- Actual: Real automated disposable recovery executed; operational configuration pending.
- Status: Blocked for business acceptance.
- Tester: automated runner where cited; business tester pending.
- Date: 2026-10-20 audit; manual date pending.
- Evidence: `verify-step11-recovery.php`; final execution logs.

UAT-10: Permissions and concurrent users

- BRD reference: 14,17,23.
- Preconditions: All six test-role users.
- Steps: Try allowed/direct denied URLs; run two cashiers on same stock/invoice.
- Expected: No unauthorized access, duplicate posting or overpayment.
- Actual: Role tests and selected native races; whole-shop manual pass pending.
- Status: Blocked for business acceptance.
- Tester: automated runner where cited; business tester pending.
- Date: 2026-10-20 audit; manual date pending.
- Evidence: `EmployeeManagementTest; verify-step12-concurrency.php`; final execution logs.

Pending-work follow-up

The earlier descriptions of missing repair stock issue and supplier/GRN services are historical and are superseded. Automated workflow cases now pass in StockPurchasingWorkflowTest (including pricing transfer

and customer advances), LatestPurchaseCostTest and PrintGapCompletionTest. Native two-user races pass in verify-pending-concurrency.php. Manual UAT statuses remain blocked/pending owner/hardware participation; they were not converted into signed business approval by automation.

Final BRD compliance report

Final native MySQL result: **218 passed (3566 assertions)**, duration 175.11s. Full evidence: `storage/logs/pending-complete-mysql.log`. The helper created its own randomized database, ran tests, rolled back/remigrated that disposable database, ran native multiuser/recovery checks, then removed only that database. Operational tables were not migrated/reset.

Focused evidence: `pending-focused-final.log`; PDF dimensions/fonts: `pending-pdf-review.json`; before/after operational schema/count evidence: `pending-schema-before.json`, `pending-schema-after.json`, `pending-preservation.log`; syntax/style evidence: `pending-syntax.log`, `pending-style-check.log`. All names are under `storage/logs/`.

The missing stock, supplier and purchase software workflows are now implemented and verified. Customer/repair/quotation/billing/POS flows integrate with the existing inventory and financial ledgers. Receipt/card, supplier/GRN and statement printing gaps have concrete PDF and history evidence. Latest purchase cost is owner-approved and applied to new transaction snapshots and valuation; existing records are preserved.

The matrix retains every original requirement ID across all 27 BRD sections. Missing operational-module rows now reference their workflow tests. No 100% compliance claim is made: owner UAT, physical devices, approved operational backup storage and unresolved purchase-cost allocation remain external limitations. Optional recommended features and excluded deployment/commercial items retain their explicit statuses.

Database audit: same schema/ERD, native constraint coverage and read-only before/after checks; no operational migrations. Financial/stock accuracy: fixture assertions plus native races, not certification of real accounts. Security: inherited route protections plus service authorization, retained audit history and existing authentication/CSRF/rate-limit tests. Performance: actual bounded local timings; no capacity claim. Printing: rendered fixture PDFs reviewed; no physical test. Backup: real disposable recovery passes; live/cloud configuration remains incomplete. UAT cases are prepared; business owner approval has not been fabricated.

See pending-completion-report.md for implemented functions and evidence, defect-register.md for remaining limitations, and modified-files.md for the full follow-up file list. Stop after this local completion work; no deployment/build/installer was run. An approved live-backup folder has not been supplied; Google Drive folder/account/OAuth fields remain blank as requested. No host/production scheduler, production build, installer, packaging or deployment was performed. Physical printer/scanner checks, realistic-volume capacity acceptance and owner signatures remain pending. Latest purchase cost was explicitly selected by the owner. Purchase-wide discount/freight allocation remains a separate unanswered decision; affected profit estimates remain incomplete rather than assuming a policy. Existing role permissions were preserved: storekeepers can receive/issue/count/adjust, while purchase approval and supplier money operations require `purchases.approve`.